

AADK 6 - Task 4 — Interactive Feature Plan & Implementation Report

Student Name: Rajat Prakash Dhal

Course/Unit: Android Development with Kotlin – Session 6

USN: 1hk22cs115

email: 1hk22cs115@hkbk.edu.in

Selected Feature Idea

Feature: Quiz App with Next, Submit, and Retry Buttons

Purpose:

The feature allows users to answer multiple-choice quiz questions interactively. Users can move to the next question, submit their answers, and retry the quiz after completion.

Target Audience:

Students or learners using a mobile app to test their knowledge in short quizzes.

1. UI Layout Sketch (Concept)



Layout Components

- **Top:** Quiz Title / Question Number
 - **Middle:** Question Text
 - **Below Question:** Multiple Choice Options (Radio buttons)
 - **Bottom Buttons:**
 - Next
 - Submit
 - Retry (visible after quiz completion)
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2. Button List & Actions

Button Action	State Modified
Next Moves to the next question	currentQuestionIndex
Submit Calculates final score and shows result	quizState = RESULT
Retry Restarts the quiz	Reset score, currentQuestionIndex, quizState

3. State Model

State	Description	Possible Transition
START	Quiz is initialized	→ QUESTION
QUESTION	User answering questions	→ NEXT / → RESULT
RESULT	Score displayed	→ RETRY
RETRY	Quiz reset	→ QUESTION

State Variables

- currentQuestionIndex
- selectedAnswer
- score
- quizState

4. Test Plan

Test Case	Button Sequence	Expected Result
Test 1	Select Answer → Next	Next question loads
Test 2	Answer all → Submit	Score screen appears
Test 3	Submit → Retry	Quiz restarts from first question
Test 4	Press Next without answer	Next button disabled or warning shown

5. Prototype / Pseudocode

state currentQuestionIndex = 0

state score = 0

state selectedAnswer = null

state quizState = "QUESTION"

IF quizState == "QUESTION"

 Display question[currentQuestionIndex]

 Display answer options

Button NEXT

 IF selectedAnswer != null

 IF selectedAnswer == correctAnswer

 score++

 currentQuestionIndex++

 selectedAnswer = null

 ENDIF

Button SUBMIT

```
quizState = "RESULT"
```

IF quizState == "RESULT"

Display score

Button RETRY

```
score = 0
```

```
currentQuestionIndex = 0
```

```
quizState = "QUESTION"
```

Advanced Case

- **Disable Next button if no option selected**

```
NextButton.enabled = selectedAnswer != null
```

- **Show Retry button only when quizState = RESULT**

IF quizState == "RESULT"

Show Retry Button

6. Reflection

When implementing this feature in Jetpack Compose, managing UI state correctly will be one of the main challenges. Multiple buttons modify different states, so maintaining a clear state model using remember or ViewModel will be important. Testing interactive flows such as rapid button clicks or incorrect answer selections will also require careful validation. Edge cases like screen rotation or activity recreation must be handled so that the quiz progress is not lost. Proper state management and UI recomposition will ensure smooth interaction.
